

Step 1: Identify the Big Problem

- The school canteen's process of serving is slow and crowded by students. Some students take too long to decide their order, the cashier manually calculates totals and change, and there is no system to monitor food supply.

Step 2: Identify three to four Sub-Problems

1. The canteen has slow serving processes and is crowded.
2. Students take too long to order their food.
3. The cashier has to manually compute totals and give change.
4. There is no system to track which food items are running out.

Step 3: Define Computational Thinking Approaches

The canteen has slow serving processes and is crowded.	Decomposition	Divide the canteen processes into smaller steps
Students take too long to order their food.	Abstraction	Display a simple digital menu where it is easily seen and accessible for everyone, along with its prices so students already know their order before proceeding to pick their food.
The cashier has to manually compute totals and give change.	Algorithm	We suggest creating a program that will send the students' order to the cashier, and automatically compute the total and their change.
There is no system to track which food items are running out.	Logical thinking	Program a system where it monitors each food item, identifying whether it's out of supply or not. Create an "alarm" which alerts the staff when it is running out. Also display the quantity of items sold and the remaining.

Step 4: Draw a flowchart or write a pseudocode for the identified sub-problem

START
CREATE order = empty list

CREATE add = "NONE"

CREATE total_price = 0

REPEAT

 OUTPUT menu_items, price_items

 INPUT student_order

 IF student_order is available THE

 ADD student_order to order

 total_price = total price + order

 ELSE

 OUTPUT "Food item is unavailable. Please choose another item."

 END IF

 INPUT "Do you want to choose another item?"

UNTIL add = "NO"

OUTPUT order, total_price

END